

İsmail Hakki Metehan Erol

Embedded Software Engineer – Agricultural & Off-Highway Systems

☎ +90 507 189 46 30 ✉ ihm.eroll@gmail.com 🔗 linkedin.com/in/ihm3 🐙 github.com/ihm3
📍 Konya, Türkiye | 🌐 Open to relocation – visa sponsorship required

Summary

Embedded Software Engineer with hands-on experience in **agricultural and off-highway machinery systems**. Specialized in **C/C++ development, CAN, J1939, and ISOBUS (ISO 11783)** communication, and **embedded Linux (Yocto/Buildroot)** platforms. Experienced in developing ECU software, HMI systems, and real-time control solutions for field machinery. Comfortable working across the full embedded stack, from low-level drivers to application-level logic, in production-oriented environments.

Technical Skills

Languages: C, C++, Embedded C, Python, Bash

Embedded Platforms: STM32, TI, ESP32, Nuvoton, Custom ECUs

Fieldbus & Protocols: ISOBUS (ISO 11783), CAN, J1939, CANopen, SPI, I2C, UART

Embedded Linux: Yocto, Buildroot, BSP Development

HMI & Visualization: Qt, LVGL, BT81X Series

Off-Highway Systems: ECU Software, RTK GNSS, IMU, Path Tracking

Quality & Debugging: MISRA-C, Bootloaders, JTAG, GDB

Tools: Git, CMake, KiCad, Jira

Professional Experience

Doken Teknoloji ve Makine

Embedded Software Engineer

2023 – Present

Konya, Türkiye

- Developed **ECU software** for agricultural machinery using **C/C++**.
- Implemented and maintained **ISOBUS (ISO 11783)** communication stacks including **Virtual Terminal (VT)** and **Task Controller (TC)**.
- Designed multi-ECU communication using **CAN and J1939** protocols.
- Built custom **embedded Linux systems** using **Yocto and Buildroot** for industrial hardware.
- Developed **HMI applications** for vehicle and implement displays using **Qt, LVGL, and BT81X**.
- Collaborated closely with hardware teams on driver development, system bring-up, and real-time performance optimization.

Lamptime Elektrik

Embedded Systems Engineer

2022 – 2023 (5 Months)

Türkiye

- Developed embedded applications involving **BLE, Zigbee**, and Raspberry Pi platforms.
- Integrated wireless communication features for IoT-enabled embedded systems.

Atiker Yazılım

Embedded Software Engineer Intern

2022

Türkiye

- Worked on **Embedded Linux** systems and **Qt-based** industrial HMI applications.
- Assisted in real-time Linux integration and system-level debugging.

Projects

- ISOBUS VT & TC Stack:** Developed ISO 11783-compliant Virtual Terminal and Task Controller implementations for agricultural ECUs.
- Smart Seeding System:** Designed embedded control software integrating motor control and GNSS for automated seeding operations.
- Off-Highway Navigation System:** Implemented RTK GNSS and IMU-based path tracking algorithms for field machinery.
- Embedded Linux BSP:** Created production-ready Yocto BSPs for custom agricultural control units.
- Vehicle HMI Dashboard:** Designed real-time dashboards for tractor and implement displays using LVGL and Qt.

Education

M.Sc. Computer Engineering | Selçuk University

2024 – Present

B.Sc. Electrical & Electronics Engineering | Sakarya University

2017 – 2022

Certifications

- ✓ Embedded Linux / Yocto / Buildroot (Udemy)
- ✓ Mastering Embedded Bare-metal Systems (Udemy)
- ✓ RTOS with FreeRTOS (Udemy)
- ✓ AUTOSAR Architecture (Udemy)
- ✓ IoT & Data Security / MQTT Server Mgmt (Udemy)
- ✓ ARM Architecture & C++ Programming (Udemy/BTK)

Languages

English: Fluent

Turkish: Native